

GOVERNMENT OF INDIA · MINISTRY OF
DEVELOPMENT OF NORTH EASTERN REGION

National Disaster Management Division · North Eastern Regional
Landslide Early Warning System

RESTRICTED OFFICIAL /
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API SPECIFICATION & INTERFACE CONTROL DOCUMENT (ICD)

AEGIS-LEWS v2.0: Comprehensive API
& Protocol Specification

RESTful Web Services, Real-Time WebSockets, and Bluetooth LE GATT Radio
Mesh Specifications

Document ID	API-ICD-NER-LEWS-2026	Release Version	2.0.4-RELEASE
System Designation	AEGIS-LEWS v2.0 (Autonomous Geotechnical Early Warning)	Release Date	September 2026
Author / Lead	AEGIS Platform API Architecture Group	Reviewed / Approved By	National Disaster Communications Standards Committee
Security Clearance	NIC-CERT-IN Level-4 · Restricted Official Distribution	Target Environments	NER 8 States EOC Command & Citizen P2P Mesh

1. API Architecture & Global Conventions

The AEGIS-LEWS API suite provides high-throughput, low-latency interoperability between Earth observation telemetry pipelines, emergency operation center command consoles, and sovereign mobile field nodes.

1.1 Base Environment URLs

- **Production Cloud (Render Live):** `https://aegis-lews.onrender.com/api`
- **Production WebSocket Bus:** `wss://aegis-lews.onrender.com/ws/mesh-bus`
- **Local Edge Development:** `http://127.0.0.1:8000/api`
- **Public Resilient Mesh Fallback:** `https://ntfy.sh/ner_landslide_alert`

1.2 Standard HTTP Response Envelope

All successful REST API responses return standard JSON payloads with ISO-8601 timestamps:

```
{
  "status": "success",
  "timestamp": 1790012469.341,
  "execution_time_ms": 14.2,
  "data": { ... }
}
```

1.3 Standard HTTP Status Codes

HTTP Status	Meaning	Condition Triggered
200 OK	Success	Telemetry retrieved, FoS model executed, or node queried successfully.
201 Created	Dispatched / Registered	BLE broadcast initiated or edge gateway registered into active mesh pool.
400 Bad Request	Validation Error	Coordinates outside monitored bounds or invalid geotechnical parameters.
401 Unauthorized	Authentication Failure	Missing or invalid EOC Officer credentials (expected: <code>prototype</code>).
503 Unavailable	Service Degraded	Cloud database down; client automatically transitions to resilient ntfy.sh fallback.

2. Core RESTful Endpoints

2.1 Weather & Orographic Precipitation Telemetry

`GET /weather` — Ingests live precipitation, atmospheric pressure, and soil saturation indices.

Parameter	Type	Required	Description
lat	Float	Yes	Latitude in decimal degrees (e.g., 25.4484 for Jaintia Hills).
lng	Float	Yes	Longitude in decimal degrees (e.g., 92.2152).

Sample Response (200 OK):

```
{
  "latitude": 25.4484,
  "longitude": 92.2152,
  "elevation_m": 1280.0,
  "current": {
    "precipitation_mm": 5.2,
    "rainfall_24h_mm": 64.8,
    "soil_moisture_index": 0.88,
    "temperature_c": 19.4,
    "wind_speed_kmh": 14.5
  },
  "status": "Within Safe Limits",
  "source": "Open-Meteo Orographic Ingestion Engine"
}
```

2.2 Geotechnical Slope Stability Evaluation

GET /risk/evaluate — Computes dynamic Factor of Safety (FoS) and slope failure probability.

Sample Request Query: /risk/evaluate?slope=52&wetness=92&insar=42.1&cohesion=22

Sample Response (200 OK):

```
{
  "factor_of_safety": 0.67,
  "failure_probability_pct": 99.4,
  "status": "CRITICAL BREACH",
  "color": "#DC2626",
  "failure_type": "Active Translational Shear",
  "estimated_impact_hours": 1.2,
  "shear_velocity_mm_day": 42.1,
  "geomechanical_metrics": {
    "driving_stress_kpa": 148.6,
    "resisting_strength_kpa": 99.5,
    "pore_pressure_ratio_ru": 0.44
  },
  "evacuation_directive": "EVACUATE IMMEDIATELY: Debris flow expected in 15 mins. Move to safe ground."
}
```

2.3 Emergency BLE Broadcast Dispatch

POST /alerts/ble-broadcast — Triggers immediate EOC hardware beacon broadcast sequence.

Sample Request Payload (JSON):

```
{
  "zone_name": "West Kameng & Tawang MCT Corridor",
  "severity": "CRITICAL",
  "risk_score": 99,
  "preset_code": 1,
  "message": "CRITICAL BREACH: Tawang Axis FoS=0.67. Active shear failure. Evacuate immediately.",
  "target_gateway": "all",
  "latitude": 27.5860,
  "longitude": 91.8650
}
```

Sample Response (201 Created):

```
{
  "status": "dispatched",
  "dispatch_id": "BLE-RELAY-E4F92A",
  "zone": "West Kameng & Tawang MCT Corridor",
  "severity": "CRITICAL",
  "risk_score": 99,
  "preset_code": 1,
  "primary_gateways_notified": 4,
  "websockets_active": 1,
  "detail": "Signal dispatched to Primary Gateway phone. BLE radio advertisement initiated for offline"
}
```

3. Real-Time WebSockets Protocol

The platform maintains a bi-directional WebSocket telemetry stream for instant synchronization between the cloud inference pipeline and connected authority dashboards:

- **Connection Endpoint:** `wss://aegis-lews.onrender.com/ws/mesh-bus`
- **Heartbeat Interval:** 15 seconds (`{ "type": "PING" }` → `{ "type": "PONG" }`)

Inbound Disaster Alert Broadcast Frame:

```
{
  "event": "DISASTER_BREACH_TRIGGER",
  "dispatch_id": "BLE-RELAY-98B10C",
  "zone": "Jaintia Hills Sector 8 (NH-44)",
  "severity": "CRITICAL",
  "fos": 0.62,
  "preset_code": 1,
  "message": "EVACUATE IMMEDIATELY: Debris flow expected in 15 mins. Move to safe ground.",
  "coords": [25.4484, 92.2152],
  "dispatched_at": "2026-09-21 23:46:14 IST"
}
```

4. Bluetooth LE (BLE) GATT Radio Interface Control

The sovereign hardware radio interface exposes standard Bluetooth SIG compliant service structures:

GATT Entity	UUID	Permissions	Description
Sovereign Service	0000AE61-0000-1000-8000-00805F9B34FB	N/A	Primary AEGIS Emergency Signaling Service (Advertised in BLE Channels 37, 38, 39).
Full Directive Char	0000AE62-0000-1000-8000-00805F9B34FB	READ (Unauthenticated)	Returns complete UTF-8 evacuation text to connecting rescue devices and civil defence radios.

4.1 Standard Preset Emergency Directive Codes

Preset Code	Title	Standard Canonical Directive Text
0x01	EVACUATE IMMEDIATELY	EVACUATE IMMEDIATELY: Debris flow expected in 15 mins. Move to safe ground.
0x02	DEBRIS FLOW IMMINENT	DEBRIS FLOW IMMINENT: Heavy rainfall detected. Stay clear of natural drainage channels.
0x03	ROAD BLOCKED	ROAD BLOCKED: Rockfall at highway corridor. Traffic suspended.
0x04	SHELTER IN PLACE	SHELTER IN PLACE: Seek designated bedrock refuge center.
0x05	FLASH FLOOD WARNING	FLASH FLOOD WARNING: Rapid runoff rising in valley floor.
0x06	BRIDGE WASHED OUT	BRIDGE WASHED OUT: Do not attempt crossing.